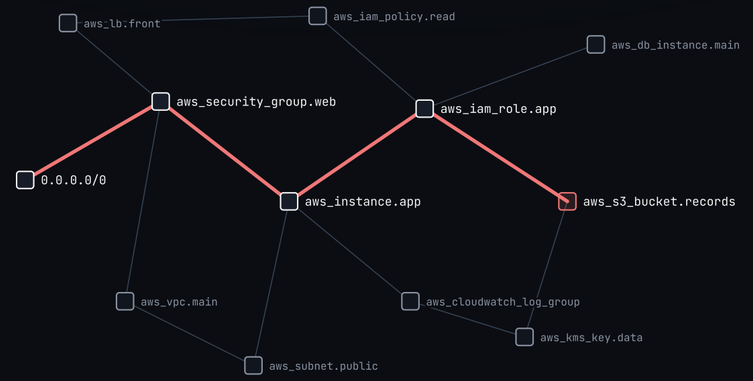
Explainable Cloud Security Analyzer with *Attack-Path* Analysis and AI-Assisted Remediation

A Terraform-Based Pre-Deployment Analysis with Optional Read-Only AWS Validation



PRESENTED BY

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SUPERVISOR

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FYP Monitoring Presentation · Tentative title

One file builds the cloud, and its mistakes.

1. Terraform declares AWS resources in code, before they exist.
2. A mistake in the file is deployed every time it is applied.
3. Scanners list mistakes one by one. This project shows how they connect, before deployment.

Detect	What is misconfigured?
Explain	Why is it risky?
Connect	How can weaknesses connect?
Prioritise	What should be fixed first?
Fix	What change resolves it?

Terraform documentation · developer.hashicorp.com ↗

main.tf · the file Checkov scans on slide 6 — open file

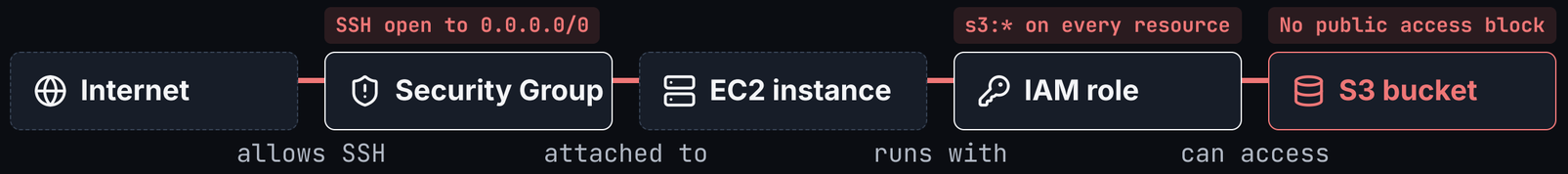
```
1  resource "aws_security_group" "web" {
2    ingress {
3      from_port   = 22
4      to_port     = 22
5      protocol    = "tcp"
6      cidr_blocks = ["0.0.0.0/0"]
7    }
8  }
9
10 resource "aws_instance" "app" {
11   ami                  = "ami-12345678"
12   instance_type        = "t3.micro"
13   vpc_security_group_ids = [aws_security_group.web.id]
14   iam_instance_profile   = aws_iam_instance_profile.app.name
15 }
16
17 resource "aws_iam_instance_profile" "app" {
18   role = aws_iam_role.app.name
19 }
20
21 resource "aws_iam_role" "app" {
22   assume_role_policy = jsonencode({ Statement = [{ Action = "sts:AssumeRole",
23     Effect = "Allow", Principal = { Service = "ec2.amazonaws.com" } }] })
24 }
25
26 resource "aws_iam_role_policy" "app" {
27   role   = aws_iam_role.app.id
28   policy = jsonencode({ Statement = [{
29     Effect = "Allow", Action = "s3:*", Resource = "*"
30   }] })
31 }
32
33 resource "aws_s3_bucket" "records" {
34   bucket = "app-records"
35 }
```

SSH open to the whole internet
`aws_security_group.web`
Why Port 22 accepts connections from any IPv4 address (0.0.0.0/0).
Fix Restrict cidr_blocks to a trusted range, or use managed access instead of SSH.

Role can do anything in S3
`aws_iam_role_policy.app`
Why Action "s3:*" on Resource "*" grants every S3 action on every bucket.
Fix Scope Action and Resource to the specific bucket and operations the app needs.

Bucket has no public access block
`aws_s3_bucket.records`
Why Nothing in the configuration prevents the bucket from being made public later.
Fix Add aws_s3_bucket_public_access_block with all four settings enabled.

Three findings. *One* potential path.



Connected: each link must be proven from the configuration. Only then is it one combined potential risk.

Detect

PS1 Detected weaknesses stay unconnected

Limited contextual analysis of interconnected IaC configurations

Scanners match known patterns one resource at a time. They rarely show how resources, identities and permissions combine into a broader risk.

Understand

PS2 The reader must work out why it matters

Limited explainability of interconnected security findings

A finding names the weakness. The reader still has to work out the evidence and why it matters.

Prioritise

PS3 Combined risk is not ranked as one

Limited contextual prioritisation of combined cloud security risks

The same weakness matters more or less depending on its surroundings. Scoring each one alone hides combined risk.

Four objectives, one analyzer.

Terraform
HCL + plan JSON

→

Rule checks
10 rules · S3 · SG · IAM

→

Relationship graph
resources · identities · permissions

→

Explain & prioritise
evidence · fix · ranking

→

CLI / HTML report
read before deploying

→

AI-assisted fix
suggest · approve · re-scan

10 rules each mapped to CIS / AWS guidance

● found in slide 2's main.tf

S3

01 Block Public Access not fully enabled ●
02 Bucket ACL or policy allows public access
03 Default encryption not configured
04 HTTPS-only access not enforced

SECURITY GROUPS

05 SSH (22) open to 0.0.0.0/0 or ::/0 ●
06 RDP (3389) open to the internet
07 All ports or protocols open to the internet

IAM

08 Wildcard Action ("*") in an Allow statement
09 Wildcard Resource ("*") for restrictable actions ●
10 Wildcard Principal in a trust or resource policy

01 Detect selected misconfigurations in Terraform

To identify and analyse selected AWS cloud misconfigurations in Terraform Infrastructure as Code using a rule-based assessment approach.

Answers PS1

02 Model relationships to find potential attack paths

To develop context-aware analysis that models relationships between selected AWS resources, identities and permissions to identify potential attack paths.

Answers PS1

03 Explain with evidence, then prioritise

To develop an explainable risk assessment approach that provides evidence-linked findings, remediation guidance and contextual risk prioritisation.

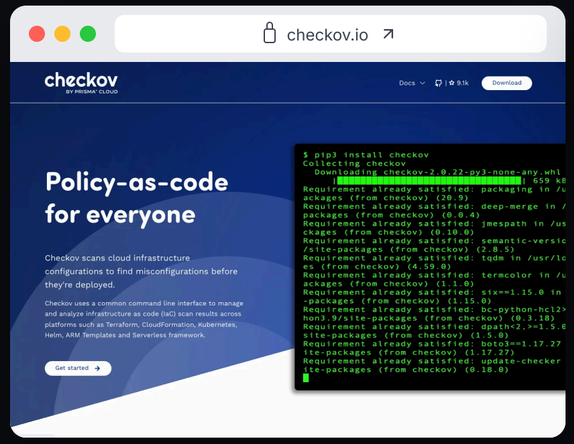
Answers PS2 · PS3

04 Evaluate on labelled scenarios against Checkov

To evaluate the proposed analyzer using labelled Terraform test scenarios, comparison with an existing IaC security scanner, and optional read-only validation against selected live AWS configurations.

8–10 labelled Terraform scenarios: ≥3 clean, 2 tricky-but-safe, ≥2 attack-path chains
Compared with Checkov on shared checks: precision, recall, F1
Optional: read-only Boto3 check of an owned sandbox account; nothing is changed

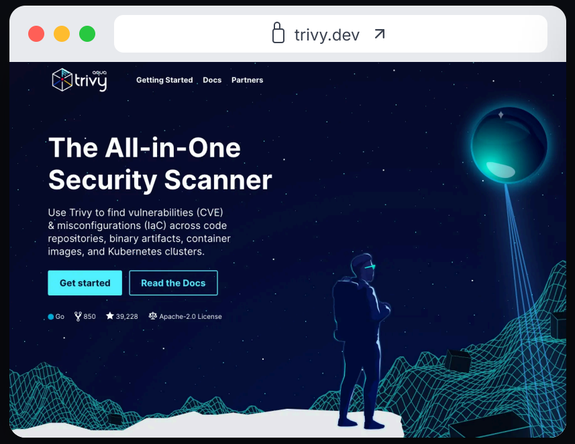
What already exists, and where this project fits.



Checkov

Scans Terraform and other IaC before deployment with built-in policies.

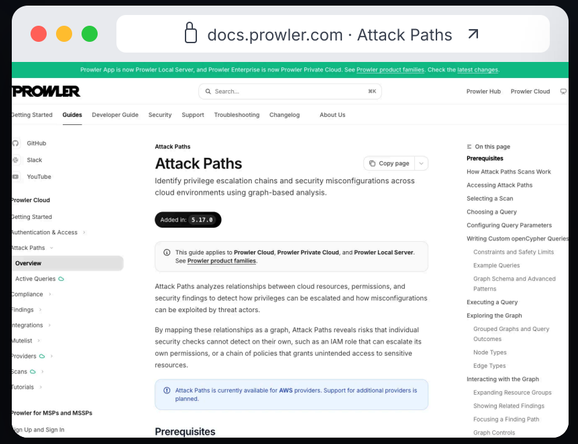
Differs: This project focuses on a small set of explained, ranked paths, measured against Checkov.



Trivy

Finds vulnerabilities, misconfigurations (including IaC) and secrets.

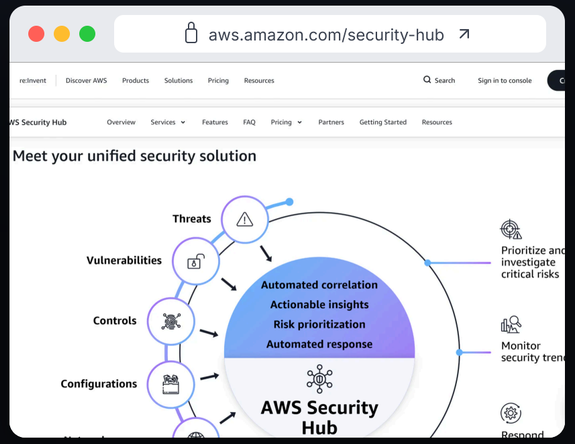
Differs: Relationship analysis was not covered in the sources reviewed; this project centres on it.



Prowler

Assesses deployed cloud accounts; Prowler App draws graph-based Attack Paths for AWS.

Differs: Works on a running account; this project works on Terraform before deployment.



AWS Security Hub

AWS's unified security service: correlates threats, vulnerabilities and configurations, and prioritises critical risks in AWS accounts.

Differs: Requires deployed resources; this project reviews the configuration first.



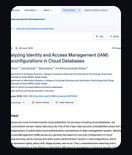
Movva, D. K., Pullikallu, A. R., & Sugasi, M. K. (2025). Infrastructure-as-Code Security and Drift Detection Using LLMs.

Reports 31% higher detection of context-dependent misconfigurations in its own evaluation: context matters.



Keerthana, K. S. (2026). Static Analysis and LLM-Based Review for Terraform Security and Cost Issues: An Exploratory Comparative Study.

Contrasts deterministic checks with contextual explanation and suggested fixes.



Almarri, A., Alawadh, A., Alsayed, M., & Rafique, M. A. (2026). Analyzing Identity and Access Management (IAM) Misconfigurations in Cloud Databases.

Excessive rights and manual policy administration lead to privilege escalation: IAM belongs in scope.

Direction Terraform detection + relationships + evidence + prioritisation, with AI-suggested fixes re-checked by the rules. Measured against Checkov.

From separate findings to one explained path.

1 Today: real scanner output

```
$ checkov -f main.tf --compact --fra…   main.tf  full output

Passed checks: 22, Failed checks: 17, Skipped checks: 0

Check: CKV_AWS_24: "Ensure no security groups allow ingress from 0.0.0.0:0 to port 22"
FAILED for resource: aws_security_group.web

Check: CKV_AWS_355: "Ensure no IAM policies documents allow "*" as a statement's resource for restrictable actions"
FAILED for resource: aws_iam_role_policy.app

Check: CKV_AWS_288: "Ensure IAM policies does not allow data exfiltration"
FAILED for resource: aws_iam_role_policy.app

Check: CKV2_AWS_6: "Ensure that S3 bucket has a Public Access block"
FAILED for resource: aws_s3_bucket.records

… and 13 other failed checks, listed one by one
```

Real Checkov 3.3.29 output on slide 2's main.tf (11 Nov 2026). Correct, but each finding stands alone.

2 The analyzer connects them

Internet — 0.0.0.0/0

allows SSH from

```
cidr_blocks = ["0.0.0.0/0"], port 22 (main.tf:3-6)
```

Security Group — aws_security_group.web

attached to

```
vpc_security_group_ids = [aws_security_group.web.id] (main.tf:13)
```

EC2 instance — aws_instance.app

runs with role

```
iam_instance_profile → aws_iam_instance_profile.app (role = aws_iam_role.app.name)
```

IAM role — aws_iam_role.app

can access

```
aws_iam_role_policy.app (role = aws_iam_role.app.id): Action = "s3:*", Resource = "*"
```

S3 bucket — aws_s3_bucket.records

3 What it would report

cloud-risk-analyzer · report.html — Planned mock-up

#1 · COMBINED RISK

Potential attack path: Internet → S3 bucket

Internet → Security Group → EC2 → IAM role → S3 bucket

WHY

An internet-reachable instance runs with a role that can read and write every S3 bucket, including app-records.

EVIDENCE

4 links, each tied to a line of main.tf

PRIORITY

Ranked above its four individual findings, as one combined risk.

FIX

Break a link: restrict SSH or detach the role. Scoping the policy limits what the role can reach.

Fix with AI → slide 7

A potential path built from configuration evidence, not a confirmed exploit.

AI-assisted *remediation*

Rules detect → AI suggests → You approve → Rules re-check

EXISTING ANALYZER · SLIDES 2-6 | EXTENSION · AI-ASSISTED FIX | OUTSIDE THE TOOL

1 YOU
Terraform
write main.tf

→

2 ANALYZER
Plan → JSON
terraform show -json

→

3 ANALYZER
Detect
rules · severity · paths

→

4 AI MODEL
AI-assisted fix
findings + main.tf → patch

→

5 YOU
Review
diff + reason · accept?

→

6 ANALYZER
Validate & re-scan
fmt · validate · same rules

→

7 YOU
Deploy
your decision

✕ reject → main.tf unchanged

4 Choose what AI should fix

report.html · Fix with AI — Planned mock-up

Fix all | Selected | By severity | Breaks a path

	FINDING	SEVERITY	BREAKS PATH
☑	**SSH open to 0.0.0.0/0** aws_security_group.web	High	Internet → S3
☑	**s3:* on every resource** aws_iam_role_policy.app	High	–
☐	**No public access block** aws_s3_bucket.records	Medium	–

AI RECEIVES — main.tf + chosen findings with line evidence
AI RETURNS — a patched main.tf + one reason per change
NEVER — AWS credentials · changes to AWS · terraform apply

5 Review the suggested diff

main.tf · 2 suggested changes

```
@@ main.tf:6 · aws_security_group.web
-     cidr_blocks = ["0.0.0.0/0"]
+     cidr_blocks = ["203.0.113.0/24"]  # admin range
```

Why SSH only from a trusted range, so the internet link of the path is gone.

```
@@ main.tf:29 · aws_iam_role_policy.app
-     Effect = "Allow", Action = "s3:*", Resource = "*"
+     Effect = "Allow", Action = ["s3:GetObject", "s3:PutObject"],
+     Resource = "${aws_s3_bucket.records.arn}/*"
```

Why Role limited to the objects of one bucket instead of every S3 action everywhere.

6 Validate & re-scan

$ after accept · same rule engine

```
$ terraform fmt -check
✓ formatted
$ terraform validate
✓ Success! The configuration is valid.
$ analyzer scan plan.json
✓ resolved  SSH open to 0.0.0.0/0
✓ resolved  s3:* on every resource
• still open  No public access block (not selected)
✓ path Internet → S3 broken
✓ 0 new findings
```

7 **Deploy?** Your call. The tool never runs `terraform apply`.

Measured findings resolved on re-scan · new findings introduced · patches passing terraform validate

Literature Keerthana (2026): LLM review suggests Terraform fixes; here every suggestion is re-checked by rules.

A community collaborator, and who else shapes the work.

COLLABORATOR · CONFIRMED

Lejiend Community

lejiend.com/community ↗ Community owner · Lejiend ↗

INPUT SOUGHT FROM THE COMMUNITY

- Practitioner feedback on the Terraform test scenarios
- Review of whether the attack-path explanations are clear
- A channel to reach cloud and technology practitioners

HOW IT IMPROVES THE PROJECT

Scenarios and explanations shaped by people who use cloud infrastructure, not only by the author.

SUPERVISOR

Dr. Fairuz Amalina Narudin

FYP supervisor · academic guidance

Input sought Scope, rule selection, evaluation design

RESPONDED · GAVE INPUT

Farhan Helmy

AI & full-stack engineer (as described on his site)

Input given Suggested the AI-assisted remediation feature (slide 7)

Input sought Feedback on the AI fix workflow and the report

farhanhelmy.com ↗ LinkedIn ↗

POTENTIAL

Wider cloud security practitioners

Industry perspectives

Input sought Common AWS misconfiguration patterns

Confirmed agreed role **Contacted** asked, response pending **Planned** outreach planned **Potential** identified only

Detect → **Connect** → **Explain** → **Prioritise** → **Fix** → **Validate**

misconfigurations in Terraform · resources, identities, permissions · evidence for every link · combined risk first · AI suggests, you approve · the same rules re-check

From configuration findings to *contextual* cloud security understanding.

Thank you. Questions?

REFERENCES

Movva, D. K., Pullikallu, A. R., & Sugasi, M. K. (2025). Infrastructure-as-Code Security and Drift Detection Using LLMs. *International Journal of Computer Security, 2(1), 9–25.* https://doi.org/10.34218/IJCST_02_01_002

Keerthana, K. S. (2026). Static Analysis and LLM-Based Review for Terraform Security and Cost Issues: An Exploratory Comparative Study. *International Journal of Engineering Applied Sciences and Technology, 11(5), 95–99.* https://doi.org/10.33564/IJEAST.2026.v11i05.010

Almarri, A., Alawadh, A., Alsayed, M., & Rafique, M. A. (2026). Analyzing Identity and Access Management (IAM) Misconfigurations in Cloud Databases. *Computers, 15(7), 409.* https://doi.org/10.3390/computers15070409

TOOLS, DOCUMENTATION & STANDARDS

Checkov · YAML custom policies (connection state)

Trivy · misconfiguration scanning

Prowler · Attack Paths

AWS Security Hub · potential attack path graph

Terraform documentation · JSON output format

CIS Amazon Web Services Foundations Benchmark v7.0.0

Lejiend Community (collaborator · community owner: linkedin.com/in/lejiend)

Screenshots of official pages captured 11 Nov 2026 for academic reference.